

ROLL NUMBER: 34

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AIM: To study AI Agents and Types of AI Agents.

OBJECTIVE:

- 1) To understand AI agents.
- 2) To study different types of AI agents and classify real-world AI agents.

THEORY:

1) What is AI agent?

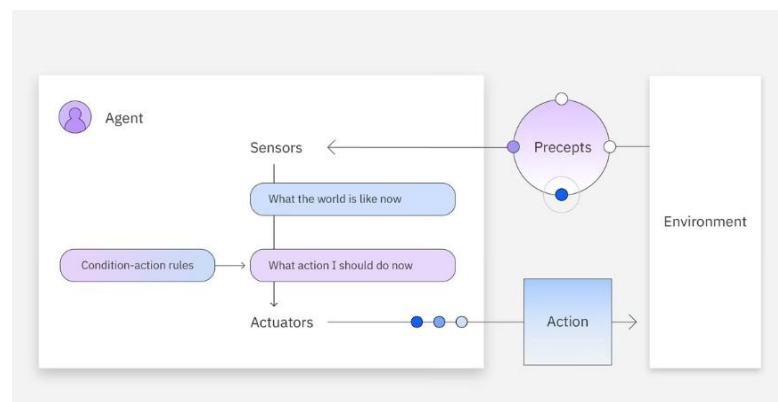
An agent is anything that can be viewed as:

- perceiving its environment through sensors and
- acting upon that environment through actuators

2) What are the 4 types of agents? Explain each type with diagram.

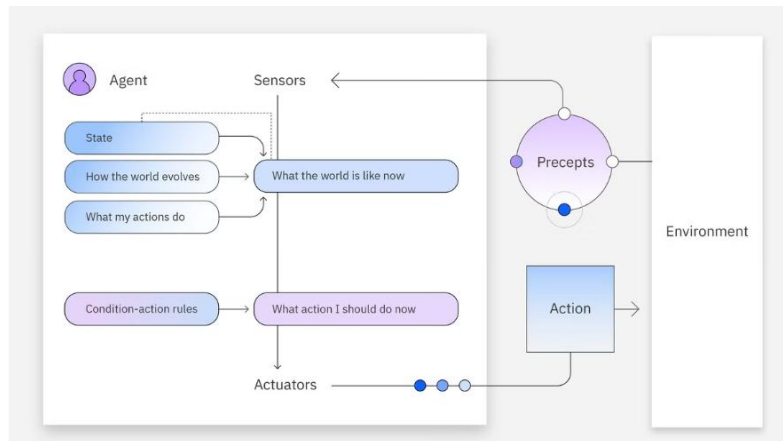
a) Simple reflex agents

A simple reflex agent is the most basic type of AI agent, designed to operate based on direct responses to environmental conditions. These agents follow predefined rules, known as condition-action rules, to determine actions without considering past experiences or future consequences.



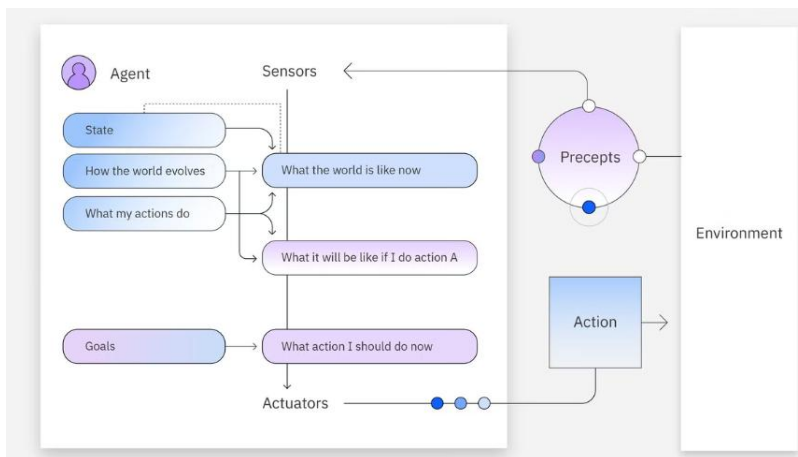
b) Model-based reflex agents

A model-based reflex agent is a more advanced version of the simple reflex agent. While it still relies on condition-action rules to decide, it also incorporates an internal model of the world. This model helps the agent track the current state of the environment and understand how past interactions might have impacted it, allowing it to make more informed decisions.



c) Goal-based agents

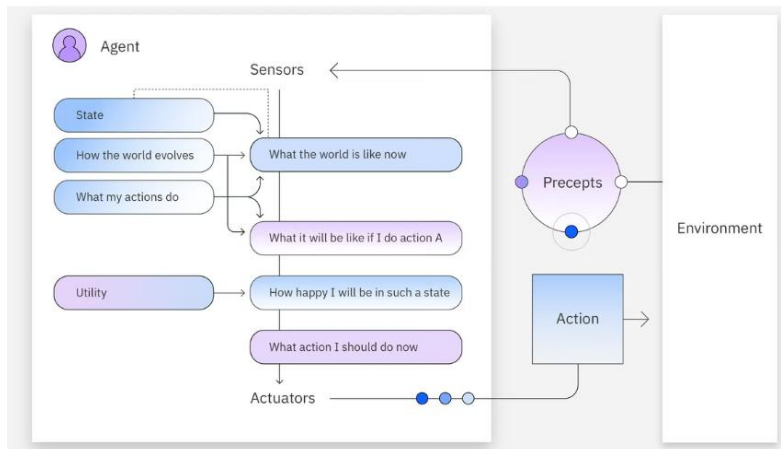
A goal-based reflex agent extends the capabilities of a simple reflex agent by incorporating a proactive, goal-oriented approach to problem-solving. These agents operate by setting a specific goal, which guides their actions. They evaluate different possible actions and select the one most likely to help them reach that goal.



d) Utility-based agents

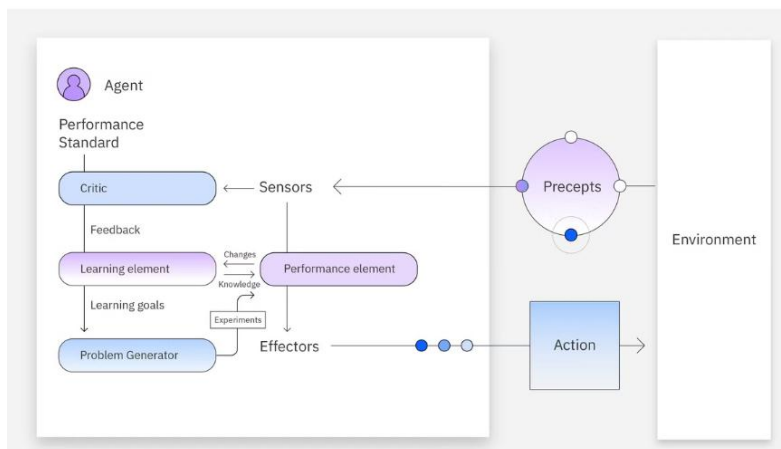
A utility-based reflex agent goes beyond simple goal achievement by using a utility function to evaluate and select actions that maximize overall benefit. While goal-based agents choose actions based on whether they fulfill a specific objective, utility-based agents consider a range of possible outcomes. They assign a utility value to each, helping them determine the optimal course of action.

This approach allows for more nuanced decision-making, particularly in situations where multiple goals or tradeoffs are involved.



e) Learning agents

A learning agent improves its performance over time by adapting to new experiences and data. Unlike other AI agents, which rely on predefined rules or models, learning agents continuously update their behaviour based on feedback from the environment.



3) Classify real-world agents (e.g., Alexa, Tesla) into reflex, goal-based, utility-based, or learning agents. [Explain at least 2 examples of each type]

1. Simple Reflex Agents

These agents react immediately to current perceptions using "if-then" rules without looking at history.

- **Thermostat:** It turns a heater on when sensors detect the temperature has dropped below a set threshold and turns it off once the desired temperature is reached.

- **Automatic Traffic Light System:** It adjusts signals based on real-time traffic sensor inputs to manage flow without referencing previous traffic patterns.

2. Model-based Reflex Agents

These agents use an internal model to track the current state and remember past interactions to handle partially observable environments.

- **Room-Navigating Robot:** Instead of just reacting to what is currently in front of it, this robot remembers the location of obstacles it has already encountered and considers its previous movements.
- **Smart Factory Maintenance Systems:** In an industrial setting, these agents track machine states over time. By maintaining this internal history of how a machine has operated, the agent can recognize maintenance needs before a failure occurs, rather than just reacting once a machine breaks down.

3. Goal-based Agents

These agents use planning and reasoning to complete specific, proactive objectives.

- **Building Navigation Robot:** A robot with the goal of reaching a specific room will plan a path that minimizes detours and avoids known obstacles, rather than just reacting to immediate stimuli.
- **XO (Tic-Tac-Toe) Agent:** As discussed in our history, an XO agent uses the goal of "getting three in a row" to guide its moves and evaluate future board states.

4. Utility-based Agents

These agents use a utility function to evaluate multiple outcomes and choose the action that provides the maximum benefit or "happiness".

- **Self-Driving Car:** When navigating, it balances competing objectives—such as speed, fuel efficiency, and passenger safety—to select the route with the highest overall utility.
- **E-commerce Pricing Agent:** These agents analyse sales history, customer preferences, and inventory levels to determine the optimal dynamic price for items.

5. Learning Agents

Learning agents improve their performance over time by adapting to feedback and new data.

- **Virtual Assistants (e.g., Alexa):** They use natural language processing (NLP) to analyse user behaviour, which allows them to optimize content recommendations and predictions based on past interactions.

- **Social Media Algorithms:** These agents analyse user interactions to predict and optimize the content shown in a user's feed, continuously refining their approach based on what the user engages with.

CONCLUSION: This assignment helped us to understand AI agents, different types of AI agents, and their real-world applications.
